MAKRAND GHODKE

M.Sc. student in Information & Communication Engineering

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Open to working student and HiWi roles in ML, AI and data | Up to 20 hours a week during the semester

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Summary

Master's student in Information and Communication Engineering at TU Darmstadt, working on machine learning and software engineering. I build and test software around ML models, mostly in Python and TypeScript. In a course project I found why an image classifier was stuck at 30% accuracy and fixed it, which took it to 100%. I later deployed that backend with Docker and added automated tests.

Projects

SpendWise, *household finance app (PWA)* 03/2026 – 05/2026
github.com/MakrandGhodke/SpendWise

- Built on my own with Next.js, TypeScript and Supabase (PostgreSQL). Row Level Security keeps each user's data separate.
- Imports bank CSV files (Revolut, N26, Sparkasse, Deutsche Bank and others) with automatic format detection, category rules and hash-based duplicate detection. Balance updates run in SQL stored procedures, so balances stay consistent.
- Works offline: changes are saved locally and synced when the connection returns. Installable as a PWA, built as an iOS app with Capacitor, deployed on Vercel, with a 6-month trend dashboard.

DigiGuide, *landmark recognition app (course project)* 10/2025 – 02/2026

- Co-developed the image classification API: CLIP embeddings and FAISS similarity search behind a FastAPI endpoint, serving live camera input for 5 landmark classes.
- Led data collection and image preprocessing. Traced the 30% accuracy to a missing BGR-to-RGB conversion; after the fix, all 5 test classes reached 100%.

DigiGuide backend: deployment and tests 10/2026
github.com/MakrandGhodke/digiguide

- Packaged the backend in Docker and deployed it on Render. The first deploy ran out of memory, so I moved the model download into the Docker build and removed the unused text encoder.
- Wrote 38 pytest tests with the model and an external API mocked, and set up GitHub Actions to run them on every push.
- The tests found three bugs: a path-traversal hole, a 500 error on invalid uploads and a hard-coded JWT secret. I fixed all three, each with its own test.

Education

B.Tech Electronics and Telecommunication, *Bharati Vidyapeeth Deemed University* 11/2020 – 07/2023
CGPA: 9.06 / 10 Pune

M.Sc. Information and Communication Engineering, *Technische Universität Darmstadt* 10/2024 – 01/01/2027
Courses: Introduction to AI, Data Science 1 and 2, Explainable and Ethical AI.

Experience

Web Developer Intern, *Exposys Data Labs* 07/2022 – 08/2022 Pune
Built and tested responsive web pages (HTML, CSS, JavaScript) in Chrome, Firefox, Safari and Edge, in an Agile team of five.

Operations Support Technician, *Flextronics Technologies* 07/2021 – 12/2021 Pune
Redesigned the Excel tracking tools used for production data reporting. Manual entry errors went down and teams could see the same data.

Skills

Machine learning
Python, PyTorch, CLIP, FAISS, scikit-learn, pandas, NumPy, computer vision, model evaluation

Backend and delivery
FastAPI, REST APIs, Docker (familiar), GitHub Actions (CI), pytest, Git, GitHub, Render, Vercel

Web and data
TypeScript, JavaScript, Next.js, React, Node.js, Flutter/Dart, SQL, PostgreSQL (Supabase), Tailwind CSS

Other
Java, C++ (bachelor's project), Jupyter Notebook, VS Code, Agile/Scrum

Languages

English
C1 (fluent)

German
A2-B1 (basic)

Hindi
native

Marathi
native